

AUTH-CAT is a package aiming to provide users with the necessary tools to quickly assess the data of an earthquake bulletin or catalogue from the Seismological Station of the Aristotle University of Thessaloniki (THE-SS).

Users can easily download an earthquake bulletin and perform a preliminary assessment through a series of Python scripts. The purpose of **AUTH-CAT** is to help students and researchers visualize the seismicity patterns of an earthquake sequence rapidly and serve as a useful tool to prepare the input of earthquake location techniques such as Hypoinverse and relocation methodologies (e.g., HypoDD).

This code is provided “as is” without any guarantees or warranty. Use at your own risk.

If you find any issue or bug, please report it in mponatis@geo.auth.gr or open an issue on the **AUTH-CAT** Github repository (https://github.com/vanagno/AUTH-CAT/issues)

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1. Installation

1.1 Install the package

You can very easily use AUTH-CAT by following these steps:

- Create and activate a dedicated environment (optional, but recommended), in order to avoid possible conflicts with packages installed in the base environment. For example, a Conda environment can be created and activated as follows:

$ conda create -n authcat

$ conda activate authcat

- Install AUTH-CAT, along with the Python libraries required (see Section 1.2), from the Python Package Index (PyPI) by running:

$ pip install authcat

- Run AUTH-CAT commands (See section 3)

1.2 Python version and required modules

AUTH-CAT was developed in Python 3.10 and is tested in Ubuntu and Windows 11. Python 3.9 or later is required.

The following Python3 modules are needed:

- Cartopy
- certifi
- corner
- emcee
- matplotlib
- numpy
- pandas
- reportlab
- Scipy
- tqdm (recommended)

The packages listed above are installed automatically together with AUTH-CAT. The installation of all required modules can be verified by running:

$ python -m authcat.requirements

1.3 Folder structure

AUTH-CAT keeps each seismicity analysis in a project folder of its own, so that different sessions are kept separately and never overwrite each other. By default, a “Projects” folder is generated and kept within the folder where AUTH-CAT is run. Its location may be changed by editing the AUTHCAT_PROJECTS environment variable. In the course of running AUTH-CAT, multiple files (.txt, .png or .pdf) are generated and archived in the folders of each project:

|— **Projects/** *(created in AUTH-CAT directory)*

|— **<name>/** *(created after running authcat bul or authcat project)*
  |— <name>.json *(project record)*
  |— **Input/**
    |— *XXXX.bul*
  |— **Results/** *(created after running authcat cat)*
    /— *XXXX.txt*
    /— *…*
  |— **Figures/** *(created after running authcat map or authcat hist, etc)*
    |— *XXXX.png*
    /— *…*
  |— **Magnitudes/** *(created after running authcat mags)*
    /— *XXXX.png*
    /— *XXXX.txt*
  |— **Reports/** *(created after running authcat rep)*
    |— <name>_report.pdf

2. Running AUTH-CAT

2.1 The authcat command

Each step of the AUTH-CAT workflow is run with the *authcat* command, followed by the name of the module

$ authcat <module command>

For example, ***$ authcat cat*** transforms the earthquake bulletin into a catalogue and ***$ authcat map*** plots the epicenters on a map. The modules are described in Section 3 and listed in Section 3.12, while ***$ authcat –help*** lists them on the screen.

AUTH-CAT modules are operated through the terminal. For modules that require additional command line options (*bul, cat, stup, pha, rep* and *project*), the available options can be displayed by running:

$ authcat <module command> -h

2.2 Projects

Each analysis is stored in a dedicated project (Section 1.3). A project can be initiated either by downloading an earthquake bulletin ***($ authcat bul***) or by creating a project from an existing bulletin with ***$ authcat project <name> –bulletin <file>***. The package currently supports bulletins in AUTOLOC3 format, with support for additional formats planned for future releases. All available projects can be listed by running ***$ authcat project***.

With the exception of ***bul***, each module first lists the projects found in the Projects folder and prompts the user to select one. If Enter is pressed without a selection, the most recently used project is selected by default. Each module then reads its input from that project and stores its output in the same project directory, so the paths given in this manual (e.g. *Results/<name> cat.txt*) are relative to the "Projects" folder. Likewise, when a module asks for a bulletin, catalogue or phase file, pressing Enter takes the project's own file (always shown in brackets) by default.

The main results produced by each module are also recorded in the project record (<name>.json) along with a history of the runs and their parameters, ensuring reproducibility. This record is also used to generate a PDF report of the project (Section 3.10).

2.3 Batch processing

There is the possibility to run all basic steps sequentially using ***$ authcat batch <project name>*** after downloading a bulletin.

This module performs all of the modules provided by AUTH-CAT except for ***rmag*** and ***pse*** since these two require judgement by the user. For the ***nt*** module, ***batch*** takes the Mc value calculated with the MAXC method in the ***mc*** step, while for the Mc over time step, it uses a 100-event window and a 50-event filter. The batch procedure produces all figures and a final report and may also be controlled in a limited way by passing arguments for mc selection and Omori-Utsu analysis period.

However, it is highly recommended to perform the steps one by one for better quality control, since an automated batch processing does not take advantage of the full potential of AUTH-CAT, given that for certain tasks, user judgement is necessary.

3. AUTH-CAT modules

A detailed overview of the modules included in AUTH-CAT is presented below. For a short description see Section 3.12. All paths are relative to the "Projects" folder.

3.1 Download the earthquake bulletin

The Central Seismological Station of the Aristotle University of Thessaloniki (THE-SS; https://seismo.auth.gr/) operating from 1981 until today, serves to provide rapid and valid information to the media and to the public, about the evaluation of seismic activity in Greece. For the daily earthquake monitoring, the SeisComp5 software is used to revise and review the original information. Earthquakes are analyzed on a 24/7 basis and final earthquake bulletins are compiled every month by the staff of THE-SS in the AUTOLOC3 format. For more detailed information see Vamvakaris & Scordilis (2020).

To download the earthquake bulletin for your area and time period of interest you need to run:

$ authcat bul

A dialogue box will appear asking for the name of the new project, and then for the start date and end date (YYYY-MM-DD) of the period of interest. Providing that valid dates are given, a message will appear asking for the minimum and maximum latitude and longitude of the study area. If geographic filtering is not selected, all earthquakes occurring within the specified time period will be retained. Finally, the user is asked to provide the name of the output file. If no filename is given, the project name is used. The resulting file will be saved to *Input/<name>.bul*.

The script downloads the monthly bulletins from the THE-SS archive (https://seismo.auth.gr/monthly-earthquake-bulletins/) and the selected events are merged into a single file. After the procedure is completed, a summary is printed indicating the number of monthly files downloaded, and the number of earthquakes written to the final bulletin.

The same can be done without any dialogue, by giving the project name, period and optionally the area as options (e.g.: ***$ authcat bul -p Arkal –start 2021-06-01 –end 2022-12-31 –min-lat 35.0 --max-lat 35.3 --min-lon 25.1 --max-lon 25.4***). You can see all available options by using ***$ authcat bul -h***.

3.2 Create and transform the earthquake catalogue

In order to create the earthquake catalogue from the bulletin downloaded from THE-SS (or any bulletin in Autoloc3 format) you need to run:

$ authcat cat

A dialogue box will appear asking for the name of the earthquake bulletin. According to the proposed folder structure, your bulletin should be in *Input/<name>.bul*. Providing that a valid bulletin is given, a message will appear asking to select the desired output format. The available options are:

i) An 11-column earthquake catalogue with columns described by its header. The generated file will be written to *Results/<name>_cat.txt*.

Date	Time	Lat	Lon	Depth	M	RMS	AzGap	Nsta	Msta	ID
2024-02-20	00:45:05.186	38.2600	20.3166	3.04	1.07	0.33	241.70	28	11	auth2024dnkq
2024-02-20	02:08:54.391	38.4114	20.4700	5.39	1.84	0.15	172.70	34	15	auth2024dnnk
2024-02-21	00:42:02.796	38.4079	20.4574	0.05	1.38	0.21	181.00	28	13	auth2024dpgb
2024-02-21	20:15:26.453	38.5444	20.6040	14.30	1.44	0.23	155.20	22	10	auth2024dqst
2024-02-22	04:00:16.775	38.5839	20.5520	12.84	1.83	0.16	176.40	26	13	auth2024dric
2024-02-22	10:50:27.738	38.2677	20.4468	0.20	0.84	0.07	255.70	8	3	auth2024drvq

Date: YYYY-MM-DD
Time: HH:mm:SS.sss
Lat & Lon: in decimal degrees
Depth: Focal depth of the event (km)
M: Earthquake magnitude with two decimals
RMS: Root Mean Square of the event (seconds)
AzGap: Azimuthal gap of the event (degrees)
Nsta: Number of total phases used for the event location
Msta: Number of seismic stations used for the magnitude calculation
ID: Unique event ID attributed by THE-SS (possibly useful for a number of tasks)

ii) A 6-column earthquake catalogue. The generated file will be written to *Results/<name>_cat_6col.txt*. According to our experience, some users prefer a more concise format of the earthquake catalogue that includes only the necessary information. This format is in accordance with the earthquake catalogue provided by THE-SS for the period 550BC - 2010 (Papazachos et al., 2000; 2010)

2024	0220004505.19	38.2600	20.3166	3.04	1.1
2024	0220020854.39	38.4114	20.4700	5.39	1.8
2024	0221004202.80	38.4079	20.4574	0.05	1.4
2024	0221201526.45	38.5444	20.6040	14.30	1.4
2024	0222040016.78	38.5839	20.5520	12.84	1.8
2024	0222105027.74	38.2677	20.4468	0.20	0.8

The newly created 6-column catalogue will include the events in the following format: YYYY MMDDHHmmSS.ss Latitude Longitude Depth Magnitude

iii) An earthquake catalogue in QuakeML format. The generated file will be written to *Results/<name>_cat.xml*. Unlike the other catalogue formats, the QuakeML file retains considerably more information from the original bulletin, including origin uncertainties, network magnitudes, seismic phase picks, phase residuals and weights, and station magnitude information. The QuakeML catalogue follows the modular XML seismological standard schema, extensively described in the QuakeML manuals (https://quake.ethz.ch/quakeml/Documents).

iv) The last option generates all catalogue formats. This is the default and strongly recommended option, since the different formats may be useful at different stages of the analysis and for further use with other seismological tools.

3.3 Plot epicentral map

After creating the earthquake catalogue, it is essential to visualize the earthquake locations. This can be done by running:

$ authcat map

A dialogue box will appear asking for the name of the earthquake catalogue. According to the proposed folder structure, it should be located in *Results/ <name>_cat.txt*. Providing that a valid catalogue is given, a map showing the epicenters included in the catalogue will be shown and a message will appear indicating that the epicentral map is saved to *Figures/Map_epic.png*.

3.4 Plotting figures and histograms

In order to assess the quality of the earthquake catalogue and make some preliminary remarks on the characteristics of seismicity, it is essential to visualize and/or quantify certain parameters. For example, by running ***$ authcat hist*** the user can generate histograms of the focal depths and magnitudes included in the catalogue, as well as the RMS and the azimuthal gaps of the solutions. For each histogram (saved to *Figures/Histograms.png*) the mean and the median of each parameter is given.

Another useful tool to assess the evolution of seismicity during a seismic excitation is a Magnitude vs Time plot which can be created with:

$ authcat mt

A dialogue box will appear asking for the name of the earthquake catalogue. According to the proposed folder structure, it should be located in *Results/ <name>_cat.txt*. Providing that a valid catalogue is given, a figure showing the events in circles (sized proportionately to their magnitude) will be shown and a message will appear indicating that the plot is saved to *Figures/M_time.png*.

Moreover, a plot showing the cumulative number of earthquakes over time is a valuable tool to reveal trends in earthquake activity over a given period, for example periods of increasing or decreasing activity. They are also useful in a preliminary identification of earthquake clusters. The user can produce this kind of plot by running:

$ authcat nt

A dialogue box will appear asking for the name of the 6-column earthquake catalogue. According to the proposed folder structure, it should be located in *Results/ <name>_cat_6col.txt*. Providing that a valid catalogue is given, another dialogue box will appear requiring the completeness magnitude threshold. This feature was added because variations in M_c during the course of a seismic excitation may significantly alter the content of this figure and lead to false conclusions. With ***$ authcat nt*** the user can test different levels of M_c and observe how they affect the cumulative number of earthquakes plot. It is recommended to use a statistically-based method to assess the M_c (see Section 3.5) and then run again this script. Ultimately, a figure consisting of 2 panels (the left panel shows the plot with all the events and the right one includes only the events above the M_c) will be shown and a message will appear indicating that the plot is saved to *Figures/cumulative_earthquakes.png.*

3.5 Magnitude of completeness analysis

AUTH-CAT provides the possibility to perform a reliable estimation of the completeness magnitude (M_c) of your earthquake catalog. Currently, AUTH-CAT supports three different ways

to assess the M_c, the maximum of curvature method (MAXC; Wiemer & Wyss, 2000), the Goodness-of-Fit method (GFT; Wiemer & Wyss, 2000) and the b-value stability (MBS; Cao & Gao, 2002) technique.

However, it has been observed that the M_c can vary significantly over time due to a number of reasons (i.e., the improvement in the detection capability). Before performing the M_c computations, it is advised to evaluate the M_c over time to gain a first order approximation and assess the variability of M_c over the selected time period. This can be achieved by running:

$ authcat mct

A dialogue box will appear asking for the name of the earthquake catalogue. According to the proposed folder structure, it should be located in *Results/<name>_cat.txt*. Providing that a valid catalogue is given, a new dialogue box will ask for the window length (number of events) for the M_c computation and the filter length (number of events) to smooth. Proposed window and filter lengths cannot be given because they are highly dependent on your data. As a rule of thumb, larger windowing reduces smaller variations and helps to see trends more clearly, however you can lose useful information, so a trial-and-error approach is recommended.

The M_c is defined as the lowest magnitude bin at which all the earthquakes in a given region are considered reliably detected. It is very crucial for a plethora of tasks such as the reliable estimation of seismicity parameters. For a more detailed explanation on the significance and the variety of approaches to estimate M_c, see the article of Pavlenko & Zavyalov (2022).

In order to assess the M_c with the methodologies supported by AUTH-CAT, you can run:

$ authcat mc

A dialogue box will appear asking for the name of the 6-column earthquake catalogue. According to the proposed folder structure, it should be located in *Results/ <name>_cat_6col.txt*. After that, a series of figures will appear showing:

i) The Frequency-Magnitude distribution (FMD) plot showing the cumulative and incremental number of events (saved to *Figures/FMD_plot.png*).

ii) The FMD plot along with the M_c estimated through the MAXC method and the b-value of the Gutenberg-Richter (G-R) relationship with its uncertainty (saved to *Figures/MAXC.png*).

iii) The GFT residuals (in %) plot at different magnitude cutoffs (saved to *Figures/ GFT_plot.png*).

iv) The FMD plot along with the M_c estimated through the GFT method (considering the 95% of residuals if possible) and the b-value of the G-R relationship with its uncertainty (saved to *Figures/GFT.png*).

v) The MBS b-value stability plot at different magnitude cutoffs (saved to *Figures/ MBS_plot.png*).

vi) The FMD plot along with the M_c estimated through the MBS method and the b-value of the G-R relationship with its uncertainty (saved to *Figures/MBS.png*).

Finally, a summary report (*Results/Mc_summary.txt*) is written with all the necessary information regarding the results of the M_c estimations.

3.6 Manipulation of magnitudes

During the routine earthquake analysis performed at THE-SS, an algorithm is employed along with the location software to determine the local magnitudes (M_L) for each event. The code uses the method of Hutton & Boore (1987) for each seismological station and calculates the final magnitude for each earthquake after averaging the individual M_L estimations.

However, in several cases it has been observed that deviations of individual M_L values in respect to the final estimations exist and they may significantly affect the reliability of the earthquake catalogue. In this respect, AUTH-CAT provides a tool perform a magnitude regularization procedure by eliminating the M_L estimations by seismic stations that systematically over- or underestimate the magnitude in comparison to the average estimations. For more information on the regularization of magnitudes, see Appendix 3 of Papadimitriou et al. (2022). To initiate the procedure, you must run:

$ authcat mags

A dialogue box will appear asking for the name of the earthquake bulletin. According to the proposed folder structure, it should be located in *Input/<name>.bul.* Providing that a valid bulletin is given, a figure consisting of two bar charts will appear showing i) the number of computed magnitudes per station and ii) the median residual between event magnitudes and individual magnitudes computed for each station. Before the figure is drawn, the user is asked for a residual threshold (e.g., 0.3). The stations whose absolute median residual exceeds this threshold, are drawn in purple color in both charts, with the threshold marked, and are listed on screen. This figure is very useful in order to assess if a seismic station is problematic and needs to be eliminated. This figure will be saved to *Magnitudes/magnitudes.png* and a list of events with their magnitudes and the individual magnitude calculations for each station will be saved to *Magnitudes/ magnitudes.txt*)

Afterwards, a dialogue box will appear asking the user if they want to recompute the magnitudes eliminating the magnitudes recorded by certain stations. If they are satisfied ("*no"),* the script will be terminated. Otherwise, ("yes") a new dialogue box will appear asking for the seismic station names to be eliminated from the magnitude calculations (pressing Enter eliminates the stations that exceed the residual threshold). After providing the station names, a new file is generated (saved to *Magnitudes/ modified_magnitudes.txt*) updating the previous file with the newly calculated magnitudes. Events left with no station magnitude after removal, keep their original magnitude. When the mean magnitudes are recomputed, a Bland-Altman plot (Bland & Altman, 1986) is also drawn and saved to *Magnitudes/bland_altman.png*). This figure is useful for the comparison between the original and the recomputed magnitudes of each event. The difference between the two is plotted against their mean, along with the mean difference and the 95% limits of agreement (mean difference ± 1.96 standard deviations) showing how much the eliminated stations influenced the newly computed magnitudes. It is highly recommended that the magnitude calculation and recalculation steps are done right after the bulletin download and transform step, in order to use the corrected magnitudes for every step of the workflow.

If the user is satisfied with the result, they can update the seismic catalogue with the newly calculated magnitudes by running:

$ authcat rmag

A dialogue box will appear asking for the name of the initial earthquake catalogue. According to the proposed folder structure, it should be located in *Results/ <name>_cat.txt*. Providing that a valid catalogue is given, a new dialogue box will appear asking for the newly calculated magnitudes (according to the proposed folder structure, it should be located in *Magnitudes/modified_magnitudes.txt*). Providing that a valid file is given, a new earthquake catalogue with the revised magnitudes is generated and saved to *Results/<name>_cat_mod.txt*.

It is possible to repeat some or all previous steps by giving the new catalogue (*Results/<name>_cat_mod.txt*) when a command asks for the earthquake catalogue. In fact, it is strongly advised, given that if the revised magnitudes highly differ from the initial ones, this may affect the M_c and subsequently the estimation of seismicity parameters.

3.7 Seismic phases extraction

Determining the number and the distribution of P- and S- phase arrivals in an earthquake dataset is crucial given that it affects the accurate location of the focal parameters and allows the user to perform a quality control of the seismic network. AUTH-CAT provides the capability to count the number of P- and S- wave arrivals at each seismic station and generate a visual representation of phases counts, by running:

$ authcat sta

A dialogue box will appear asking for the name of the earthquake bulletin. According to the proposed folder structure, it should be located in *Input/ <name>.bul*. Providing that a valid bulletin is given, a map of the seismic stations will appear, color-coded according to their total number of recorded P-phases. This map is saved to *Figures/Map_Stations.png*. Simultaneously, a list of the number of P- and S-phases recorded for each station is saved to *Results/station_phases_counts.txt*. Stations missing from the station coordinates file are listed on screen. The module ***$ authcat stup*** can look them up and add them to the file (Section 3.11).

In case the user wants to create a list of seismic phases in HYPO71 or HypoDD format, this can be done by running:

$ authcat pha

A dialogue box will appear asking for the name of the earthquake bulletin. According to the proposed folder structure, it should be located in *Input/ <name>.bul*. Providing that a valid bulletin is given, the user chooses the format of the phase file: HYPO71 (written to *Results/phases.txt*), HypoDD (written to *Results/phase.dat*, the phase input of ph2dt), or both (default).

However, usually we do not want to keep all the seismic phases, for example phases from very distant stations, or bad quality phases. The user can select which seismic station phase picks they want to keep by running:

$ authcat pse

A dialogue box will appear asking for the list of seismic phases generated from the previous step. According to the proposed folder structure, it should be located either in *Results/phases.txt* or *Results/phase.dat*. Then, a new dialogue box will appear asking the user to enter the station codes they want to keep (separated by commas). The number of P and S phases of each station is listed first, to help the choice, and pressing Enter by default (without providing station names), keeps the 20 stations with the most P phases. The updated list of seismic phases in HYPO71 or HypoDD format is then saved to *Results/phases_sel.txt* or *Results/phases_sel.dat*.

3.8 V_p/V_s estimation

Defining the ratio of the velocity of P-waves to the velocity of S-waves is a crucial parameter particularly in the context of earthquake location. Along with crustal velocity models, it is essential to interpret the seismic wave travel times and help to identify regional velocity variations, leading to more precise earthquake locations. In this context, the Wadati method is the most common technique to estimate V_p/V_s by analyzing the travel times of P-waves and S-waves from a series of earthquakes. This can be achieved by running:

$ authcat vpvs

A dialogue box will appear asking for the name of the earthquake bulletin. According to the proposed folder structure, it should be located in *Input/<name>.bul*. Providing that a valid file is given, the Wadati plot will appear showing the P-wave travel times (x-axis) against the time separation between the S-wave and P-wave travel times (y-axis). AUTH-CAT supports two regression techniques (Least squares fitting and orthogonal regression) to estimate V_p/V_s. The plot is saved to *Figures/Wadati.png*.

3.9 Omori-Utsu law parameters

The Omori-Utsu (OU) law describes the decay rate of aftershock activity following a main earthquake. It is represented by $n(t)=K*(t+c)^{-p}$, where n(t) is the rate of aftershocks at time t after the main shock, K is the parameter describing the productivity of aftershocks, c is a time constant that adjusts the starting time of the aftershock sequence (due to the short-term aftershock incompleteness) and p is the decay rate exponent that essentially determines how quickly the aftershock rate decreases over time. The reliable estimation and understanding of the OU law can help predict the frequency and intensity of aftershocks following a strong earthquake, which is crucial for assessing ongoing risks and planning for disaster response. The determination of the OU parameters can be performed by running:

$ authcat ou

A dialogue box will appear asking for the name of the earthquake catalogue. According to the proposed folder structure, it should be located in *Results/<name>_cat.txt*. Providing that a valid catalogue is given, the user chooses the start of the aftershock sequence (the start of the catalogue, the largest earthquake, or a specific date and time), its end date (by default, the end of the catalogue) and, optionally, a magnitude threshold (e.g., the M_c). The Omori–Utsu parameters are first estimated by maximizing the likelihood of the observed event times. These maximum-likelihood estimates are used as the initial parameter estimates for the fit. Parameter uncertainties are then assessed within a Bayesian framework using Markov chain Monte Carlo (MCMC; Goodman & Weare, 2010) sampling with *emcee* (Foreman-Mackey et al., 2013), incorporating the adopted prior distributions. The resulting posterior distributions are used to derive median parameter values and credible intervals.

Two figures are shown: the aftershock rate over time with the fitted OU law (saved to *Figures/Omori.png*), and the distributions of the sampled parameters, drawn with corner (Foreman-Mackey, 2016) saved to *Figures/Omori_MCMC.png*). The binned rates and the fitted values are saved to *Results/Omori_fit_data.txt*.

3.10 Summary report

After completing the desired AUTH-CAT steps, the results can be summarized in a single PDF report by running:

$ authcat rep

During the execution of the AUTH-CAT modules, the main information and results produced at each step are stored in the project record, a JSON file located in the "Projects" folder (*<name>.json).* The script reads the analysis summary stored in the JSON file and gathers the available catalogue information, numerical results and figures into a structured PDF report. Only analyses for which results are available are included in the final report. The generated report is saved in *Reports/<name>_report.pdf.*

3.11 Station list update

The coordinates of the seismic stations, used by ***$ authcat sta*** (Section 3.7), are listed in the station file *regd.sta (Figure 1)*, which is installed with AUTH-CAT and shared by all projects. As seismic networks change over time, with stations installed, moved or closed, the station list can be updated by running:

$ authcat stup

The command finds every data node of the European Integrated Data Archive (EIDA) through the EIDA routing service, and then asks their FDSN station web services, and that of IRIS, for the stations within Greece and its surrounding region (default: latitude 33.5-43.5°N, longitude 18-31°E. Another area can be given with --region). Each station already in the list takes its most recent position, and every station of the area missing from the list is added, while no station is ever removed. A station whose new position lies more than 10km from the listed one is reported and its new position is not written in the file, unless –accept-far is given. A summary of the changes is printed, and the updated list is written, keeping the previous one as *regd.sta.bak*.

3.12 Brief overview of AUTH-CAT modules

Name	Description	Required Input	Output
bul bulletin.py	Download the earthquake bulletin from THE-SS	Project name	Earthquake bulletin (.bul)
cat catalog.py	Create the earthquake catalogue from the bulletin. Transform into 6-column and XML	Earthquake bulletin (.bul)	Earthquake catalogue (.txt)
map seismicity_map.py	Plot the epicenters in the catalogue	Earthquake catalogue (.txt)	Epicentral map (.png)
hist histograms.py	Plot histograms of focal depths, magnitudes, RMS and azimuthal gaps	Earthquake catalogue (.txt)	Figure (.png)
mt M_time.py	Plot the magnitudes over time	Earthquake catalogue (.txt)	Figure (.png)
nt N_time.py	Plot the cumulative number of earthquakes over time	Earthquake catalogue (.txt)	Figure (.png)
mct Mc_time.py	Plot the M_c variations over time	Earthquake catalogue (.txt)	Figure (.png)
mc Mc_analysis.py	Determine the M_c of the earthquake dataset	Earthquake catalogue (.txt)	Figures (.png)
mags magnitudes.py	Plot the median residual of magnitudes for each station	Earthquake bulletin (.bul) & station codes to remove from magnitude estimation (optional)	Figure (.png) & List of magnitudes per seismic station (.txt)

rmag repmags.py	Replace the initial magnitude estimations	Earthquake catalogue (.txt) & newly computed magnitudes (.txt)	Earthquake catalogue (.txt)
sta stations.py	Visualize the seismic stations with the number of picked phases	Earthquake bulletin (.bul) & station coordinates (regd,sta)	List of stations and phases counts (.txt) & map of seismic stations (.png)
stup update_stations.py	Update the station list from the FDSN station services	Station list (regd.sta)	Updated station list (regd.sta)
pha phases.py	Extract seismic phases from the bulletin	Earthquake bulletin (.bul)	List of seismic phases in HYPO71 or/and HypoDD format (.txt)
pse sel_phases.py	Select only the phases from certain seismic stations	Phases list (.txt)	Selected phases list (.txt)
vpvs wadati.py	V_p/V_s estimation through the Wadati method	Earthquake bulletin (.bul)	Figure (.png)
ou omori_utsu.py	Estimation of the Omori-Utsu parameters	Earthquake catalogue (.txt)	Figure (.png)
rep report.py	Generate summary report	Summary file (.json)	Report (.pdf)
project project.py	List the projects, or start one from a bulletin you already have	Earthquake bulletin (.bul)	Project folder & project record (.json)

4. Example

In order to illustrate the functionality of AUTH-CAT, we used an example dataset from the 27 September 2021 M_w5.9 Arkalochori (central Crete) seismic sequence in central Crete (Greece). The dataset includes the intense foreshock activity that began in early June 2021, the mainshock, and the subsequent aftershock sequence. A step-by-step walkthrough is presented below:

$: Command run by the user
Command screen output
User input

The coordinates of seismic stations across Greece and the surrounding region are stored in the regd.sta station file (Figure 1), which is installed alongside AUTH-CAT and shared by all projects (Section 3.11). As a result, the earthquake bulletin is normally the only input required. If one or more stations are missing from the file, or if their coordinates need to be updated, the stup command can query FDSN station services and update the local station list accordingly.

```
AAMF 37.1070 22.0689
ABEA 35.5163 24.0113
ACHL 39.0083 22.9581
ACOR 37.8902 22.8692
AGB  35.1403 25.0022
AGG  39.0211 22.3360
AGL  35.0957 24.6882
AGN1 35.1879 25.7155
AGR2 38.6319 21.4136
AGR3 38.5893 21.4162
AIA1 40.1641 21.8202
AIG2 38.2417 22.0726
AIGS 38.2500 22.0878
AKR  36.3567 25.3917
AKR1 38.1536 22.3133
AKRO 36.3584 25.3967
AKS  38.8790 27.8130
ALIK 38.2605 22.1113
ALIO 38.0657 23.6727
ALNA 39.1477 23.8667
ALX2 40.8454 25.8739
AMGA 36.8316 25.8938
AMOR 36.9898 25.9863
AMPE 38.3015 23.2688
AMPK 39.0574 26.3130
AMPL 38.9223 21.2135
AMT  37.5324 21.7089
ANAF 36.3567 25.7727
ANKY 35.8670 23.3012
```

Figure 1: Example of the file containing the list of seismic station coordinates. Each entry identifies a seismic station and its geographic coordinates.

- Project creation and bulletin download

We start a project, named "Arkalochori", by downloading the earthquake bulletin from THE-SS for the period from 1 June to 31 December 2021 and the area between 35.08°N and 35.24°N and between 25.15°E and 25.36°E. Since the project name, the period and the area are given as options, no questions are asked (Section 3.1):

$ authcat bul -p Arkalochori --start 2021-06-01 --end 2021-12-31 --min-lat 35.08 --max-lat 35.24 --min-lon 25.15 --max-lon 25.36

```
[1/7] 2021_06.bul  ok (downloaded; 16 events matched)
[2/7] 2021_07.bul  ok (downloaded; 88 events matched)
[3/7] 2021_08.bul  ok (downloaded; 95 events matched)
[4/7] 2021_09.bul  ok (downloaded; 391 events matched)
[5/7] 2021_10.bul  ok (downloaded; 534 events matched)
[6/7] 2021_11.bul  ok (downloaded; 118 events matched)
[7/7] 2021_12.bul  ok (downloaded; 89 events matched)

Wrote Input/Arkalochori.bul (5322295 bytes)
  monthly files: downloaded 7   from cache 0   missing 0   failed 0
  events examined: 5375
  events in date range: 5375
  geographic bounds: lat 35.08000..35.24000, lon 25.15000..25.36000
  events written: 1331
```

Out of the 5375 earthquakes in the seven monthly bulletins, the 1331 that occurred in the study area are written to the project's bulletin, Input/Arkalochori.bul (Figure 2).

```
Event:
    Public ID             auth2021kupw     1st event in the bulletin
    Preferred Origin ID   NLL.20210628093253.514773.25127
    Preferred Magnitude ID Magnitude/20210628093308.191828.25135
    Description
      region name: East Crete Isl. - S. Greece
    Creation time         2021-06-03 14:06:05

Origin:
    Public ID             NLL.20210628093253.514773.25127
    Date                  2021-06-03
    Time                  14:05:28.788
    Latitude               35.14559 deg  +/-    2.540 km
    Longitude              25.25903 deg  +/-    1.002 km
    Depth                     0.098 km   +/-    5.563 km
    Agency                AUTH
    Author                username@nano
    Mode                  manual
    Status                confirmed
    Creation time         2021-06-28 09:32:55
    Residual RMS              0.195 s
    Azimuthal gap             153.5 deg

2 Network magnitudes:
    MLh       2.29 +/- 0.20   7 preferred  AUTH
    M         2.29            7            AUTH

13 Phase arrivals:
    sta   net      dist   azi  phase   time            res     wt  sta
    KNSS  HC    0.17128 337.9  P       14:05:32.278  -0.068 M  1.4  KNSS
    KNSS  HC    0.17128 337.9  S       14:05:35.215   0.093 M  1.4  KNSS
    NPS   HL    0.30971  67.9  P       14:05:34.864  -0.046 M  1.4  NPS
    NPS   HL    0.30971  67.9  S       14:05:39.904   0.219 M  1.1  NPS
    IDI   HL    0.33362 295.4  P       14:05:34.592  -0.853 M  0.2  IDI
    IDI   HL    0.33362 295.4  S       14:05:39.639  -0.999 M  0.2  IDI
    .......
    .......

7 Station magnitudes:
    sta   net      dist   azi  type   value   res        amp  per
    KNSS  HC    0.17128 337.9  MLh     2.48  0.18    2.98231
    NPS   HL    0.30971  67.9  MLh     2.10 -0.20   0.627369
    IDI   HL    0.33362 295.4  MLh     2.14 -0.15   0.626493
    .......
    .......
                End of 1st event
                Start of the 2nd event
Event:
    Public ID             auth2021kvlq
    Preferred Origin ID   NLL.20210604051500.396872.23144
    Preferred Magnitude ID Magnitude/20210604051518.074569.23183
    Description
      region name: East Crete Isl. - S. Greece
    Creation time         2021-06-04 01:05:52
```

Figure 2: Example of the AUTOLOC3 earthquake bulletin.

- Catalogue creation

Every command begins by prompting the user to select which project to work on. For clarity, this is omitted from the command-line outputs shown below.
To create the earthquake catalogue from the bulletin we run:

$ authcat cat

Input folder:
  Arkalochori.bul
Enter the name of the earthquake bulletin file [blank = Arkalochori.bul]: Arkalochori.bul

Output formats:
  1 classic 11-column catalogue (_cat.txt)
  2 6col 6-column analyst format (_cat_6col.txt)
  3 quakeml QuakeML 1.2 (_cat.xml)
  4 all all three [default]
Which format? 4

Read 1331 events, 2641 magnitudes, 35139 phase arrivals, 17323 station magnitudes from Input/Arkalochori.bul
Results/Arkalochori_cat.txt: 1331 events
Results/Arkalochori_cat_6col.txt: 1331 events, 6 origin times carried into the next second when rounded to 1/100 s, 138 origin times exactly halfway between centiseconds, rounded up
Results/Arkalochori_cat.xml: 1331 events, 35139 picks, 17323 station magnitudes

Date	Time	Lat	Lon	Depth	M	RMS	AzGap	Nsta	Msta	ID
2021-06-03	14:05:28.788	35.1456	25.2590	0.10	2.29	0.20	153.50	13	7	auth2021kupw
2021-06-04	01:05:26.207	35.1623	25.2485	13.13	4.19	0.44	149.70	62	34	auth2021kvlq
2021-06-04	01:58:45.675	35.1440	25.2698	5.78	2.85	0.22	154.30	30	15	auth2021kvnk
2021-06-05	23:49:00.713	35.1842	25.2559	5.49	1.98	0.04	145.90	16	6	auth2021kzac
2021-06-06	13:16:39.249	35.1623	25.2651	8.82	2.47	0.42	173.20	11	5	auth2021laat
2021-06-08	16:14:59.792	35.1505	25.2688	7.54	2.82	0.26	153.00	31	18	auth2021ldxq
2021-06-08	16:29:52.339	35.1594	25.2659	10.48	2.55	0.30	151.10	25	10	auth2021ldyd
2021-06-08	16:35:16.951	35.1521	25.2678	9.50	3.15	0.32	152.60	37	21	auth2021ldyi
2021-06-08	16:38:17.126	35.1578	25.2532	8.33	3.12	0.39	150.80	36	20	auth2021ldyk
2021-06-08	17:54:39.081	35.1562	25.2717	10.09	2.71	0.32	152.00	28	14	auth2021leay
2021-06-08	21:25:00.906	35.1448	25.2644	8.13	2.48	0.30	153.90	23	12	auth2021lehw
2021-06-09	02:36:28.677	35.1753	25.2456	11.95	3.82	0.23	167.40	18	9	auth2021lesd
2021-06-12	22:47:44.526	35.1586	25.2600	11.46	3.30	0.47	151.00	79	52	auth2021llsp
2021-06-15	12:22:20.059	35.1480	25.2566	7.94	3.42	0.23	152.90	32	12	auth2021lqkm
2021-06-23	20:22:49.878	35.1383	25.2669	0.10	2.69	0.16	155.30	22	13	auth2021mfqj

Figure 3: Example of the 11-column earthquake catalogue generated from the bulletin.

Now, three new files are created and saved to the <Results> folder: the 11-column catalogue Arkalochori_cat.txt (Figure 3), the 6-column catalogue Arkalochori_cat_6col.txt (Figure 4) and the QuakeML catalogue Arkalochori_cat.xml.

2021	0603140528.79	35.1456	25.2590	0.10	2.3
2021	0604010526.21	35.1623	25.2485	13.13	4.2
2021	0604015845.68	35.1440	25.2698	5.78	2.9
2021	0605234900.71	35.1842	25.2559	5.49	2.0
2021	0606131639.25	35.1623	25.2651	8.82	2.5
2021	0608161459.79	35.1505	25.2688	7.54	2.8
2021	0608162952.34	35.1594	25.2659	10.48	2.5
2021	0608163516.95	35.1521	25.2678	9.50	3.1
2021	0608163817.13	35.1578	25.2532	8.33	3.1
2021	0608175439.08	35.1562	25.2717	10.09	2.7
2021	0608212500.91	35.1448	25.2644	8.13	2.5
2021	0609023628.68	35.1753	25.2456	11.95	3.8
2021	0612224744.53	35.1586	25.2600	11.46	3.3
2021	0615122220.06	35.1480	25.2566	7.94	3.4
2021	0623202249.88	35.1383	25.2669	0.10	2.7
2021	0627221939.56	35.1781	25.2322	13.03	2.5

Figure 4: Example of the 6-column earthquake catalogue.

- Plot epicentral map

We proceed to the visualization of the epicenters in the catalogue by running:

$ authcat map

Results folder:
Arkalochori_cat_6col.txt
Arkalochori_cat.txt
Arkalochori_cat.xml
Please enter the name of the earthquake catalogue [blank = Arkalochori_cat.txt]:
Arkalochori_cat.txt
Epicentral map saved to Figures/Map_epic.png

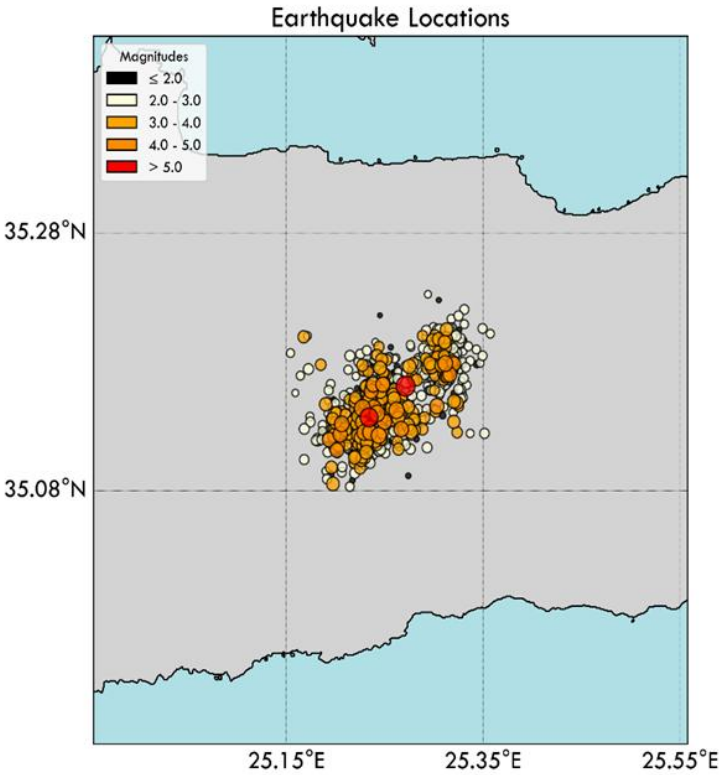


Figure 5: Map of the earthquake locations. Symbol size and color vary with earthquake magnitude.

As shown in Figure 5, a map is created with circles of different sizes and colors according to the earthquake magnitude in the catalogue. Now we have a first order approximation of where the events are located and how clustered they are in space. In this example, the epicentral distribution defines a relatively compact elongated zone in central Crete, trending approximately NE-SW.

- Plot histograms

Attempting to gain more information on the characteristics of our dataset, we run:

$ authcat hist

Results folder:
Arkalochori_cat_6col.txt
Arkalochori_cat.txt
Arkalochori_cat.xml
Please enter the name of the earthquake catalogue [blank = Arkalochori_cat.txt]:
Arkalochori_cat.txt
Histograms saved to Figures/Histograms.png

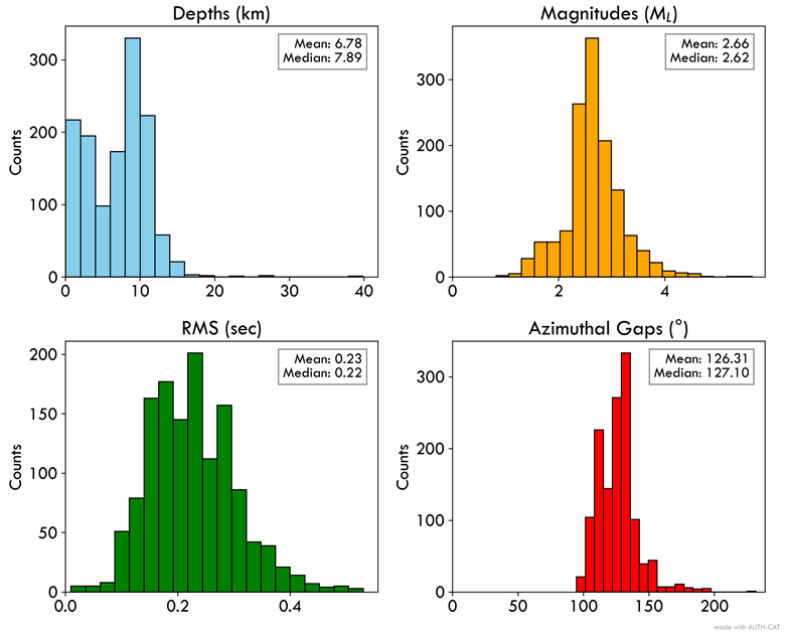


Figure 6: Histograms of focal depths, earthquake magnitudes, RMS residuals and azimuthal gaps for the catalogue.

The generated histograms (Figure 6) contain useful information especially if our goal is to proceed to earthquake relocation at later stages.

- Plot magnitudes over time

Next, we also want to gain insight on the temporal progression of magnitudes. This can be easily done by running:

```
$ authcat mt

Results folder:
Arkalochori_cat_6col.txt
Arkalochori_cat.txt
Arkalochori_cat.xml
Please enter the name of the earthquake catalogue [blank = Arkalochori_cat.txt]:
Arkalochori_cat.txt
Magnitudes vs time plot saved to Figures/M_time.png
```

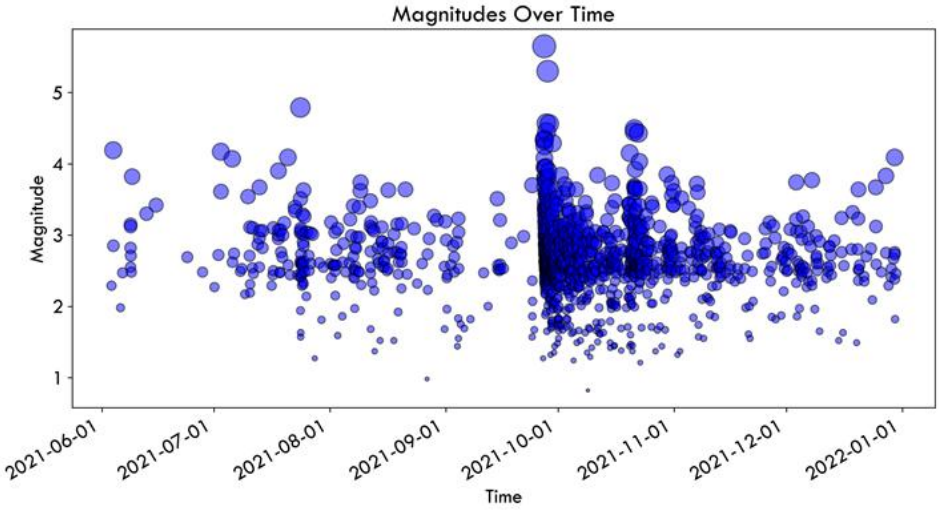


Figure 7: Magnitudes vs time plot. Symbol size is proportional to earthquake magnitude.

The resulting plot (Figure 7) indicates the onset of the intense pre-seismic activity, depicts the implied variations of M_c and a first order approximation of the total aftershock duration. In our case, the foreshock activity that started in June 2021 is followed by the main shock of 27 September 2021 and its aftershocks, which continue until the end of the year.

- Cumulative number of earthquakes over time plot

Another useful plot, complementary to the one generated from the previous step, is that of the cumulative number of events over time. This can be performed by running:

$ authcat nt

```
Results folder:
Arkalochori_cat_6col.txt
Arkalochori_cat.txt
Arkalochori_cat.xml
Please provide the name of the 6-column catalogue [blank = Arkalochori_cat_6col.txt]:
Arkalochori_cat_6col.txt
Give the Mc (Magnitude of Completeness): 2.5
2 rows are out of chronological order; sorting by time before counting
1331 events, 984 at or above Mc 2.5, over 209.7 days
Cumulative earthquakes plot saved to Figures/cumulative_earthquakes.png
```

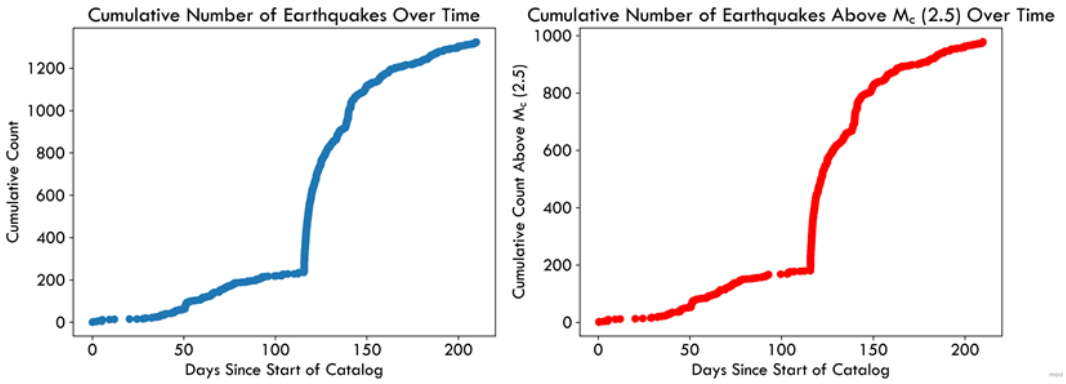


Figure 8: Cumulative number of earthquakes through time plots.

The generated plot shows the cumulative number of all events (left) and of those above the M_c given (right). We chose to give a M_c=2.5 given that this is the threshold that THE-SS provides in the revised earthquake catalogues. However, it is recommended to repeat this step after estimating the M_c statistically.

- Magnitude of completeness over time plot

Next, before estimating the M_c with the provided methods, it is very useful to gain insight to the variations of M_c with time. This can be done by running:

$ authcat mct

```
Results folder:
Arkalochori_cat_6col.txt
Arkalochori_cat.txt
Arkalochori_cat.xml
Please enter the name of the earthquake catalogue [blank = Arkalochori_cat.txt]:
Arkalochori_cat.txt
Enter the window length (no of events) for Mc computation: 50
Enter the filter length (no of events) for smoothing: 150
```

Mc vs time plot saved to Figures/Mc_time.png

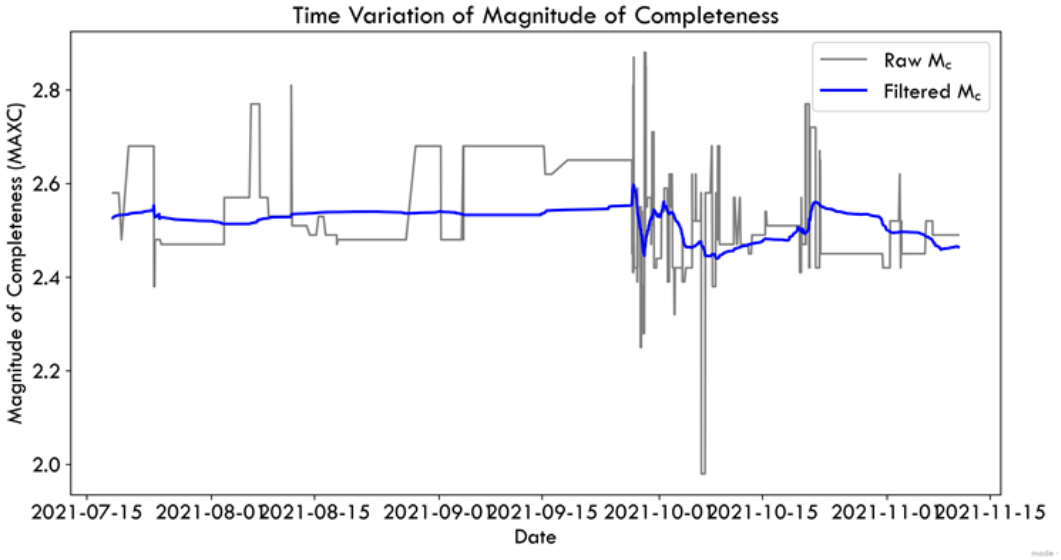


Figure 9: Temporal variation of the magnitude of completeness estimated with the MAXC method using a moving window and a smoothing filter.

After providing the earthquake catalog, based on the total number of events in the catalogue, we chose to set the window length to 50 events and the filter length to 150 (Figure 9). Other values may also be appropriate, depending on the properties of the dataset and the user's judgment. In this case, the filtered M_c stays between about 2.45 and 2.55 throughout the period.

- Magnitude of completeness and seismicity parameters determination

Now we may estimate the M_c and the seismicity parameters of our dataset. This can be done by running:

$ authcat mc

Results folder:
Arkalochori_cat_6col.txt
Arkalochori_cat.txt
Arkalochori_cat.xml
Please enter the file path of the 6-column earthquake catalogue [blank =
Arkalochori_cat_6col.txt]: Arkalochori_cat_6col.txt
Frequency-Magnitude distribution plot saved to Figures/FMD_plot.png
MAXC plot saved to Figures/MAXC.png
Goodness-of-Fit residual plot saved to Figures/GFT_plot.png
Goodness-of-Fit plot saved to Figures/GFT.png
b-value stability plot saved to Figures/MBS_plot.png
MBS plot saved to Figures/MBS.png
Summary of Mc analysis saved to Results/Mc_summary.txt

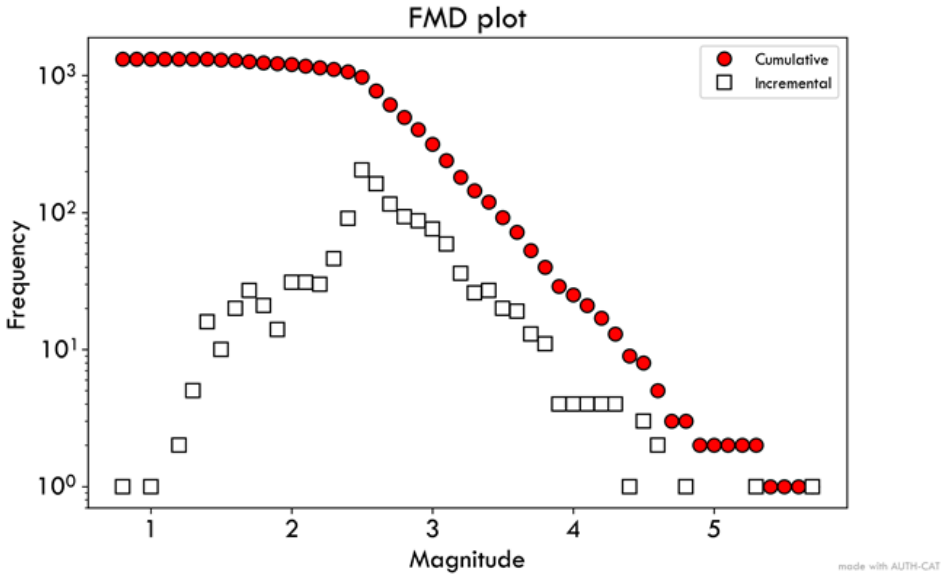


Figure 10: Frequency-Magnitude distribution of the events in the seismic catalogue.

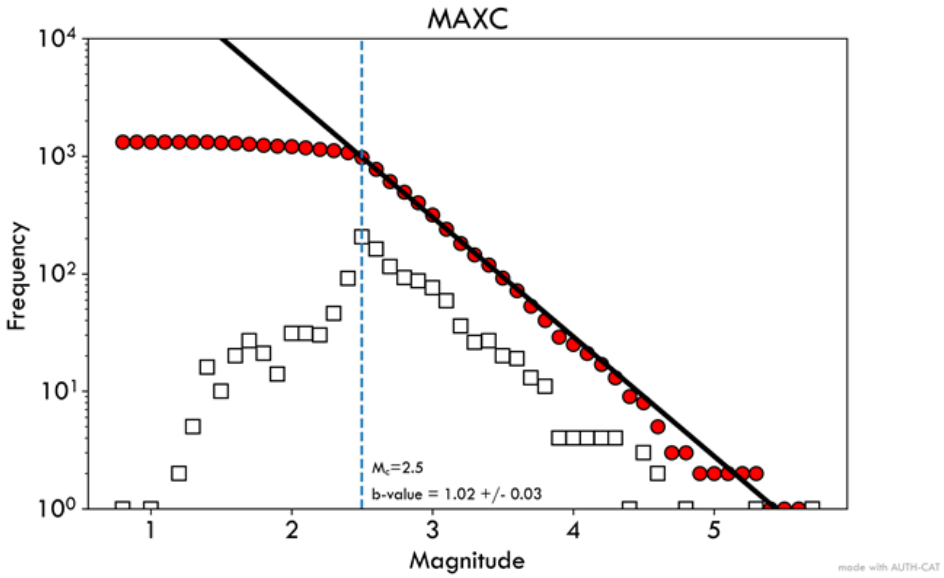


Figure 11: M_c determination using the MAXC method.

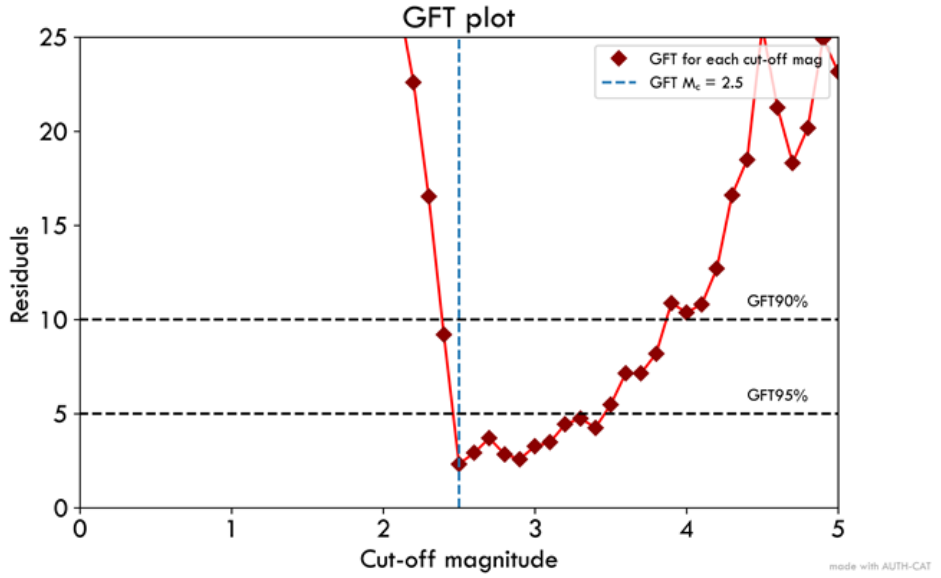


Figure 12: GFT residual plot. Horizontal dashed lines indicate the 90% and 95% goodness-of-fit thresholds, and the vertical dashed line marks the selected completeness magnitude.

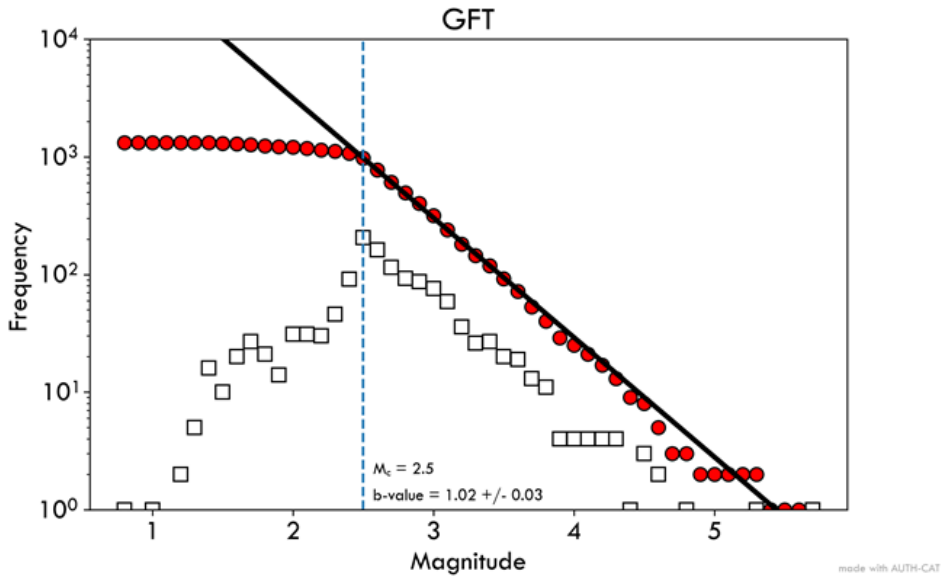


Figure 13: M_c determination using the GFT method.

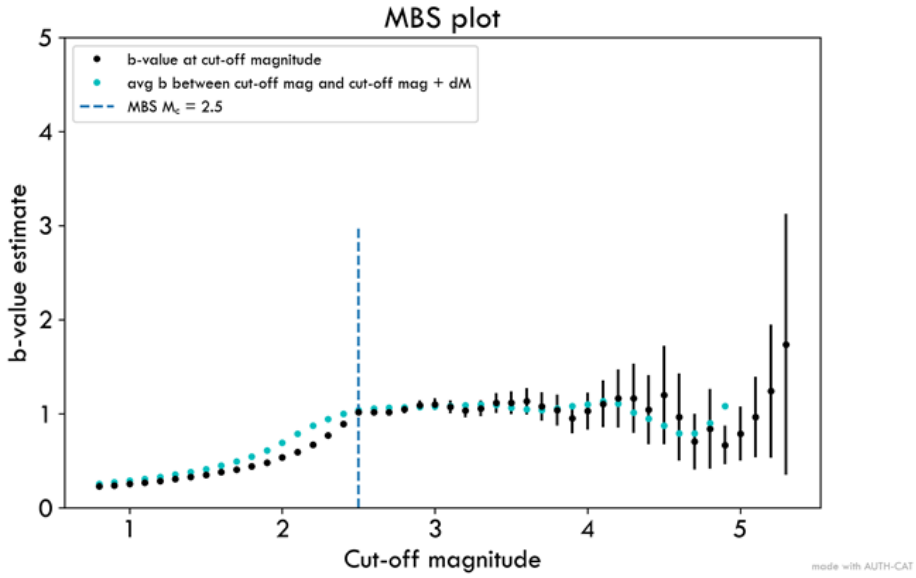


Figure 14: Stability of b-value plot for the MBS method. The vertical dashed line indicates the selected completeness magnitude.

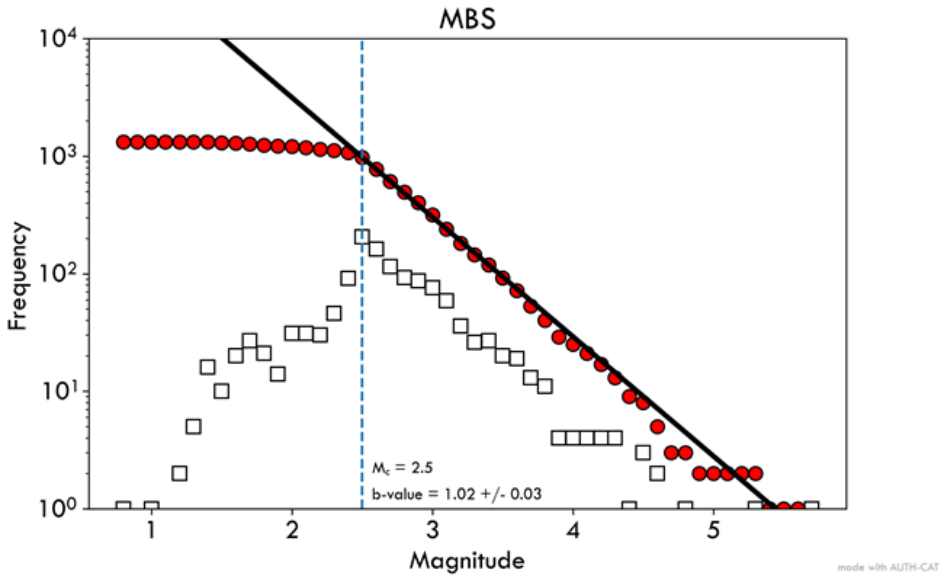


Figure 15: M_c determination using the MBS method.

A variety of plots are generated (Figures 10-15) which help to effectively select the optimal value of M_c and the G-R parameters. In our case, all three methods give M_c = 2.5 and a b-value of 1.02 ± 0.03.

- Manipulation of magnitudes

In order to check if certain seismic stations systematically over- or underestimate the magnitude in comparison to the average estimations, we run:

$ authcat mags

Input folder:
Arkalochori.bul
Enter the path to the earthquake bulletin file [blank = Arkalochori.bul]: Arkalochori.bul

Residual threshold: highlight stations whose abs median residual exceeds (e.g. 0.3, blank = none): 0.4
Median residual magnitudes plot saved to Magnitudes/magnitudes.png
Stations with abs median residual > 0.4: IACM (-0.43), SAP3 (-0.55), MHLO (-0.41)
Do you want to recompute magnitudes? (yes/no): yes
Which station magnitudes do you want to eliminate? (with commas) [blank = IACM, SAP3, MHLO]: IACM, SAP3, MHLO
Created a modified output file with specified stations removed.
Updated the output file by removing specified stations.
Do you want to compute mean magnitudes for each event? (yes/no): yes
Mean magnitudes computed and appended to the file.
New mean magnitudes were computed and written to Magnitudes/modified_magnitudes.txt
Bland-Altman plot saved to Magnitudes/bland_altman.png

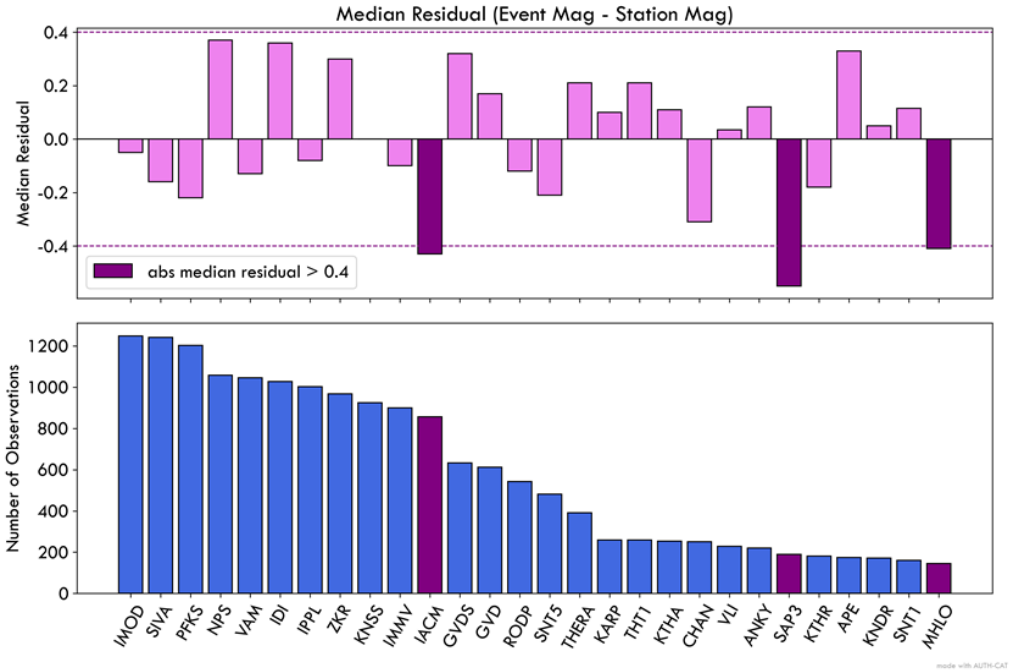


Figure 16: Median residual of magnitudes plot. Stations exceeding the pre-defined residual threshold are colored in deep purple.

After providing the earthquake bulletin and a residual threshold of 0.4, a plot is generated (Figure 16). As we can observe, IACM, SAP3 and MHLO overestimate the magnitudes of the events (median residual below -0.4, drawn in purple). As a result, we chose to eliminate them from magnitude calculations. The criteria upon which certain stations could be eliminated or not are not standard, so it is up to the user's judgement. Thus, the script computed new magnitudes without the aforementioned stations and created a new file (Figure 17) which includes the initially calculated ones as well as the new ones for comparison purposes.

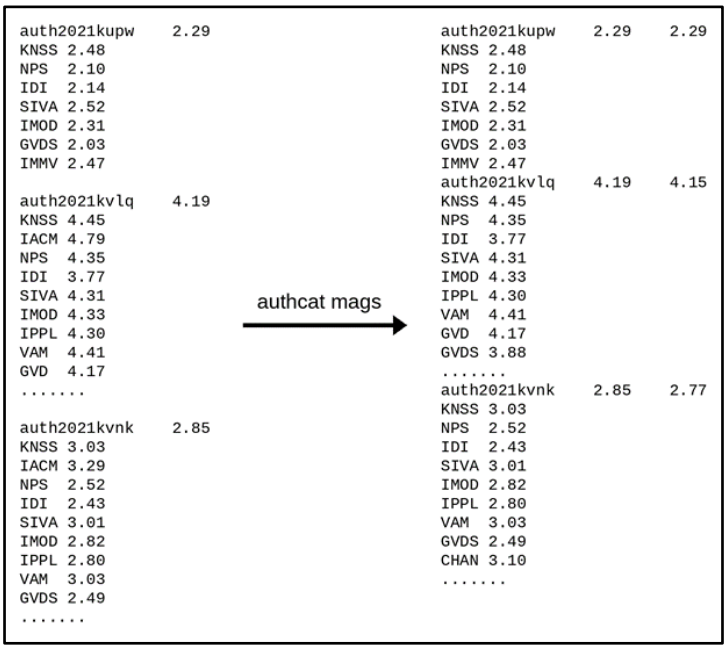


Figure 17: Example of the file with the recomputed magnitudes (modified_magnitudes.txt).

Since the magnitudes were recomputed, a Bland-Altman plot compares the original and recomputed magnitudes of the 1331 events (Figure 18). On average, the recomputed magnitudes are 0.03 lower than the original ones, and 95% of the differences lie between -0.085 and 0.025, as expected after leaving out three stations that overestimate the magnitudes.

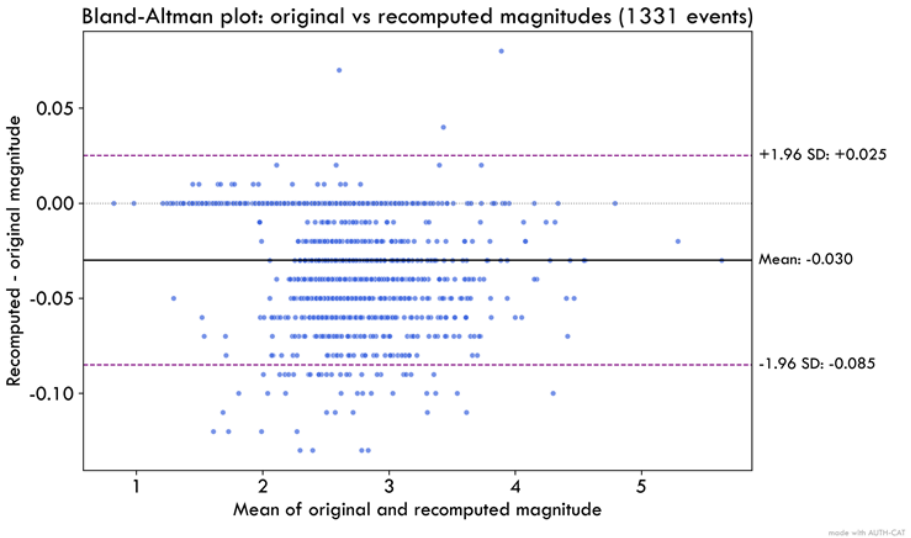


Figure 18: Bland-Altman plot of the original and recomputed magnitudes.

Given that we are satisfied with the newly calculated magnitudes, we want to create the earthquake catalogue with the "corrected" magnitudes. This can be done by running:

$ authcat rmag

Results folder:
Arkalochori_cat_6col.txt
Arkalochori_cat.txt
Mc_summary.txt
Arkalochori_cat.xml

Please provide the name of the initial catalogue [blank = Arkalochori_cat.txt]:
Arkalochori_cat.txt
Magnitudes folder:
modified_magnitudes.txt
magnitudes.png
magnitudes.txt
bland_altman.png
Great! Now provide the modified mean magnitudes file
(should be modified_magnitudes.txt if not renamed) [blank = modified_magnitudes.txt]:
modified_magnitudes.txt
Catalog has been updated with modified magnitudes and saved to
Results/Arkalochori_cat_mod.txt

```
   Date        Time       Lat     Lon   Depth  M   RMS  AzGap  Nsta Msta    ID
2021-06-03 14:05:28.788 35.1456 25.2590  0.10 2.29 0.20 153.50  13   7 auth2021kupw
2021-06-04 01:05:26.207 35.1623 25.2485 13.13 4.19 0.44 149.70  62  34 auth2021kvlq
2021-06-04 01:58:45.675 35.1440 25.2698  5.78 2.85 0.22 154.30  30  15 auth2021kvnk
2021-06-05 23:49:00.713 35.1842 25.2559  5.49 1.98 0.04 145.90  16   6 auth2021kzac
2021-06-06 13:16:39.249 35.1623 25.2651  8.82 2.47 0.42 173.20  11   5 auth2021laat
2021-06-08 16:14:59.792 35.1505 25.2688  7.54 2.82 0.26 153.00  31  18 auth2021ldxq
2021-06-08 16:29:52.339 35.1594 25.2659 10.48 2.55 0.30 151.10  25  10 auth2021ldyd
2021-06-08 16:35:16.951 35.1521 25.2678  9.50 3.15 0.32 152.60  37  21 auth2021ldyi
2021-06-08 16:38:17.126 35.1578 25.2532  8.33 3.12 0.39 150.80  36  20 auth2021ldyk
2021-06-08 17:54:39.081 35.1562 25.2717 10.09 2.71 0.32 152.00  28  14 auth2021leay

                          authcat rmag
                                ↓

   Date        Time       Lat     Lon   Depth  M   RMS  AzGap  Nsta Msta    ID
2021-06-03 14:05:28.788 35.1456 25.2590  0.10 2.29 0.20 153.50  13   7 auth2021kupw
2021-06-04 01:05:26.207 35.1623 25.2485 13.13 4.15 0.44 149.70  62  34 auth2021kvlq
2021-06-04 01:58:45.675 35.1440 25.2698  5.78 2.77 0.22 154.30  30  15 auth2021kvnk
2021-06-05 23:49:00.713 35.1842 25.2559  5.49 1.97 0.04 145.90  16   6 auth2021kzac
2021-06-06 13:16:39.249 35.1623 25.2651  8.82 2.41 0.42 173.20  11   5 auth2021laat
2021-06-08 16:14:59.792 35.1505 25.2688  7.54 2.78 0.26 153.00  31  18 auth2021ldxq
2021-06-08 16:29:52.339 35.1594 25.2659 10.48 2.52 0.30 151.10  25  10 auth2021ldyd
2021-06-08 16:35:16.951 35.1521 25.2678  9.50 3.11 0.32 152.60  37  21 auth2021ldyi
2021-06-08 17:54:39.081 35.1562 25.2717 10.09 2.67 0.32 152.00  28  14 auth2021leay
```

Figure 19: Example of the catalogue with the recomputed magnitudes (Arkalochori_cat_mod.txt)

- Seismic phases extraction

Given that our main goal is to help researchers conduct their own research, we also provide certain tools to extract and manipulate the seismic phases included in the earthquake bulletin (i.e., in order to perform earthquake relocation). To visualize the seismic stations that recorded the events in our dataset, we run:

$ authcat sta

Input folder:
Arkalochori.bul
Please provide the name of the earthquake bulletin [blank = Arkalochori.bul]: Arkalochori.bul
List of the number of P- and S-phases recorded for each station is saved to
Results/station_phases_counts.txt
Map of the seismic stations used is saved to Figures/Map_Stations.png

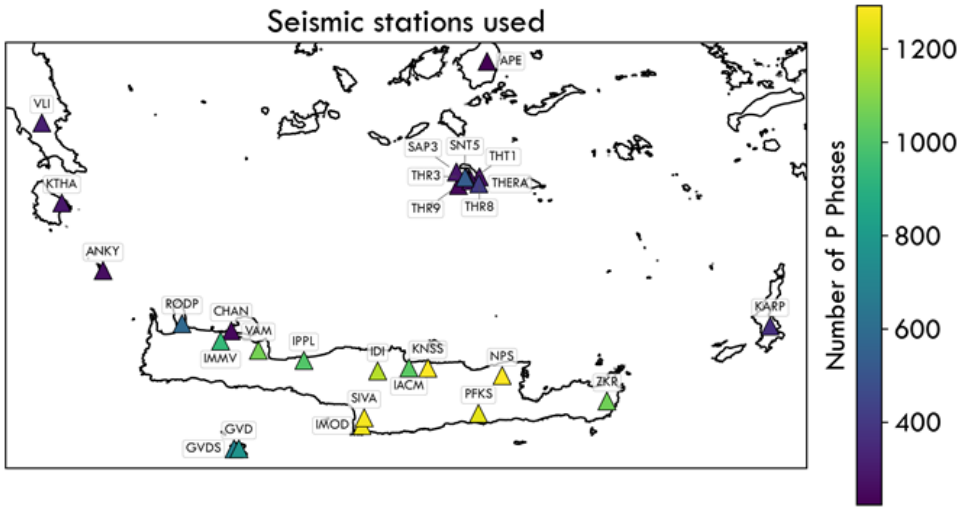


Figure 20: Map of the seismic stations used for the location of the events in the catalogue. Stations are colour-coded according to the number of recorded P-wave arrivals.

After providing the earthquake bulletin, the script warns us if some of the seismic stations are not included in the station coordinates file (in our case, all 133 stations are included). Missing stations can be looked up and added to the station file with $ authcat stup (Section 3.11). A map is generated (Figure 20) which gives us a very good image of which stations contribute most to the located events. Moreover, a list of the seismic stations with the corresponding P- and S-phases is written (Figure 21).

KNSS	35.3042	25.1802	1293	1217
NPS	35.2613	25.6103	1290	1259
SIVA	35.0175	24.8100	1279	1230
IMOD	34.9731	24.7995	1269	1220
PFKS	35.0430	25.4733	1249	1226
IDI	35.2880	24.8900	1174	1136
VAM	35.4068	24.1997	1070	929
ZKR	35.1147	26.2170	1059	939
IACM	35.3058	25.0708	1019	940
IPPL	35.3487	24.4632	1004	967
IMMV	35.4607	23.9812	940	596
GVD	34.8391	24.0874	772	370
GVDS	34.8389	24.0585	688	292
RODP	35.5604	23.7577	582	312
SNT5	36.4037	25.3953	534	208

Figure 21: Example of the list of P- and S-phases counts.

In parallel, we can create a file of the recorded seismic phases in HYPO71 format (Figure 22) and HypoDD formats by running:

$ authcat pha

Input folder:
Arkalochori.bul
Please provide the name of the earthquake bulletin [blank = Arkalochori.bul]: Arkalochori.bul
Read 1331 events and 35139 picks from Input/Arkalochori.bul

Which phase format(s) should be written?
1) hypodd -> phase.dat HypoDD / ph2dt phase input
2) hypo71 -> phases.txt HYPO71 phase list

3) both
Please choose 1, 2 or 3 [default both]: 3
HypoDD phase input -> Results/phase.dat: 1331 events, 35132 phases written; 7 phases with travel time <= 0
HYPO71 phase list -> Results/phases.txt: 1331 events, 34905 phases written; 16 S picks too late to fit the HYPO71 seconds field (>= 100 s after P); their P is written alone; 206 S picks with no P at the same station (a HYPO71 card takes its date and minute from the P)

```
GVDSEP   210603140548.13
IDI EP   210603140534.59       39.64 S 2
IMMVEP   210603140549.44
IMODEP   210603140536.80
KNSSEP   210603140532.28       35.21 S 2
NPS EP   210603140534.86       39.90 S 2
SIVAEP   210603140536.43       42.42 S 2
THR9EP   210603140551.85
ZKR EP   210603140543.69
                   10
AMT EP   210604010625.47
ANKYEP   210604010556.16
APE EP   210604010557.73
ARG EP   210604010608.59
CHANEP   210604010546.22       60.27 S 2
CHOSEP   210604010617.66
CMBOEP   210604010549.69
DIONEP   210604010615.38
EPIDEP   210604010613.79
GUR EP   210604010622.63
GVD EP   210604010545.06       58.42 S 2
GVDSEP   210604010546.02       60.06 S 2
IACMEP   210604010531.01       34.71 S 2
IDI EP   210604010532.20       37.58 S 2
.......
```

Figure 22: Example of the seismic phases list (in HYPO71 format).

However, when performing earthquake relocation, it is customary not to use all the seismic phases. Usually, seismic station recordings up to a certain distance or a threshold number of phases are kept. In order to select which seismic phases you want to keep you must run:

$ authcat pse

Results folder:
station_phases_counts.txt
Arkalochori_cat_6col.txt
phases.txt
Arkalochori_cat.txt
Mc_summary.txt
Arkalochori_cat.xml
Arkalochori_cat_mod.txt
phase.dat
Please provide the phases list (should be phases.txt for HYPO71 or phase.dat for HYPODD formats if not renamed) [blank = phases.txt]: phases.txt

Read 1331 events and 20776 cards (34905 P and S phases) from 133 stations (HYPO71 format)

Phases per station in Results/phases.txt:

STA	LAT	LON	P	S	TOTAL
KNSS	35.3042	25.1802	1293	1211	2504
NPS	35.2613	25.6103	1290	1252	2542
SIVA	35.0175	24.8100	1277	1222	2499

```
 IMOD  34.9731  24.7995      1268   1210   2478
 PFKS  35.0430  25.4733      1249   1215   2464
 IDI   35.2880  24.8900      1174   1130   2304
……
……
Enter the station codes you want to keep (separated by commas) [blank = the 20 with the
most P phases]:
Keeping the 20 stations with the most P phases: KNSS, NPS, SIVA, IMOD, PFKS, IDI, VAM, ZKR,
IACM, IPPL, IMMV, GVD, GVDS, RODP, SNT5, THER, KARP, THR3, VLI, KTHA
Kept 16961 of 20776 cards in 1331 of 1331 events
Phase list has been successfully updated to Results/phases_sel.txt
```

```
GVDSEP   210603140548.13
IDI EP   210603140534.59       39.64 S 2
IMMVEP   210603140549.44
IMODEP   210603140536.80
KNSSEP   210603140532.28       35.21 S 2
NPS EP   210603140534.86       39.90 S 2
SIVAEP   210603140536.43       42.42 S 2
ZKR EP   210603140543.69
                10
GVD EP   210604010545.06       58.42 S 2
GVDSEP   210604010546.02       60.06 S 2
IACMEP   210604010531.01       34.71 S 2
IDI EP   210604010532.20       37.58 S 2
IMMVEP   210604010546.52       61.29 S 2
IMODEP   210604010534.25       41.18 S 2
IPPLEP   210604010538.47       48.91 S 2
KARPEP   210604010554.59
.......
```

Figure 23: Example of the updated list of phases based on the selected seismic stations from the user.

By pressing Enter, we kept the seismic phases recorded by the 20 stations with the most P phases (Figure 23).

- V_p/V_s estimation

In order to estimate the V_p/V_s through the Wadati method you have to run:

```
$ authcat vpvs

Input folder:
Arkalochori.bul
Please enter the name of the earthquake bulletin file [blank = Arkalochori.bul]: Arkalochori.bul
Left out 4 P-S pairs whose P arrives before the origin time
Wadati diagram saved to Figures/Wadati.png
```

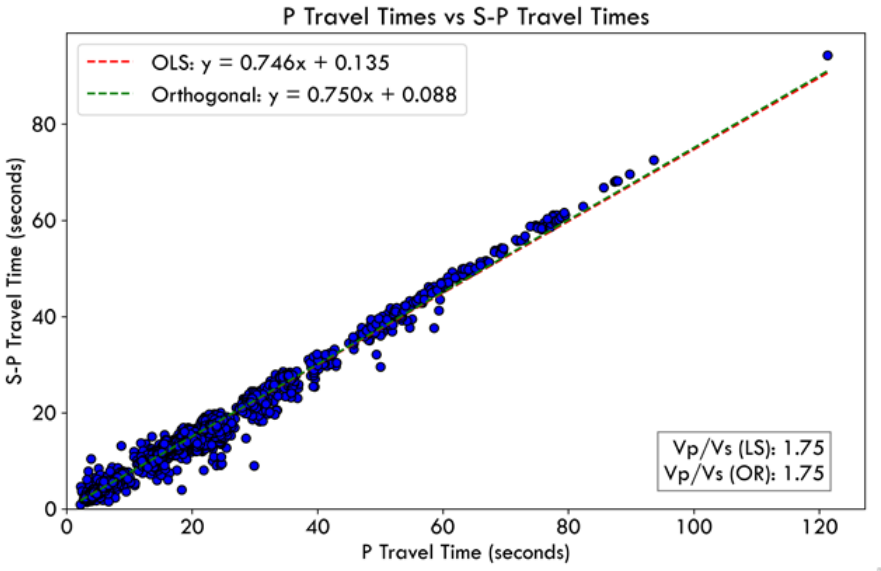


Figure 24: Wadati diagram showing travel time difference as a function of P-wave travel time.

A plot was generated (Figure 24), indicating that the V_p/V_s is equal to 1.75 with both the Least Squares fitting and Orthogonal Regression. The diagram was based on 14140 pairs of P- and S-wave travel times.

- Omori-Utsu parameters estimation

Next, you can estimate the Omori-Utsu law parameters by running:

$ authcat ou

```
Results folder:
  Arkalochori_cat.txt
  Arkalochori_cat_6col.txt
  Arkalochori_cat_mod.txt
  Mc_summary.txt
  phases.txt
  phases_sel.txt
  station_phases_counts.txt

Please enter the earthquake catalogue [blank = Arkalochori_cat.txt]: Arkalochori_cat.txt

Loaded 1331 events from Results/Arkalochori_cat.txt.
Catalogue time span: 2021-06-03 14:05:28.788000 to 2021-12-30 07:14:44.407000

Select the starting point of the aftershock sequence:
  1 - Start of catalogue
  2 - Largest earthquake
  3 - Specific date/time
Choice [1/2/3]: 2

Largest earthquake:
  Time:  2021-09-27 06:17:21.314000
  Magnitude: 5.65
  Event ID:  auth2021sxyu
```

```
Enter end date/time [YYYY-MM-DD HH:MM:SS] (press Enter for the end of the catalogue):
Ending at catalogue end: 2021-12-30 07:14:44.407000

Enter magnitude threshold Mc (press Enter to use all events): 2.5

Omori-Utsu analysis selection:
  Start mode:   largest_event
  Start time:   2021-09-27 06:17:21.314000
  End time:     2021-12-30 07:14:44.407000 (end of catalogue)
  Magnitude:    M >= 2.5
  Events used:  734
  Time span:    0.00338725 to 93.7415 days

Estimating Bayesian Omori-Utsu parameters...

Bayesian Omori-Utsu parameters:
  K = 80.0881
  c = 0.121954
  p = 0.796397

Running MCMC uncertainty analysis...

MCMC parameter estimates (median, 16th-84th percentile):
  c = 0.146529 (+0.0779 / -0.0529)
  K = 82.9843 (+9.05 / -7.2)
  p = 0.807954 (+0.0359 / -0.0327)

Outputs:
  Omori-Utsu plot:    Figures/Omori.png
  MCMC corner plot:   Figures/Omori_MCMC.png
  Fit data:           Results/Omori_fit_data.txt
```

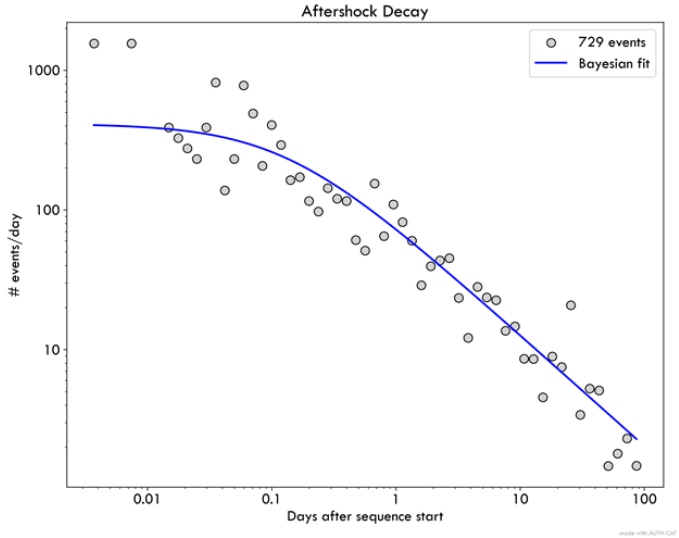


Figure 25: Temporal decay of the seismic sequence using the Omori-Utsu law.

Two plots were generated: the decay of the aftershock rate with the Bayesian fit of the Omori-Utsu law (Figure 25), with K = 80.09, c = 0.12 days and p = 0.80, and the posterior distributions

of the parameters from the MCMC sampling (Figure 26), whose medians (16th to 84th percentile) are K = 82.98 (75.78 to 92.03), c = 0.15 (0.09 to 0.22) days and p = 0.81 (0.78 to 0.84). The fit uses the 734 events of magnitude 2.5 or larger that followed the mainshock.

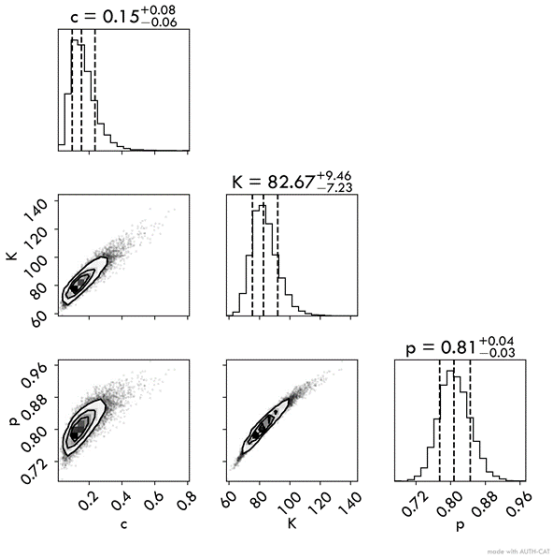


Figure 26: Posterior distributions and corner plots of the Omori-Utsu parameters derived from the MCMC analysis.

- Summary report

Finally, the results of all the steps above are gathered in a PDF report by running:

$ authcat rep

AUTH-CAT PDF report written to: Reports/Arkalochori_report.pdf

The report contains an overview of the dataset (e.g. its duration, 209.7 days, its average rate, 6.35 events/day, and its largest event) and the figures and numerical results of every step, as recorded in the project record (Arkalochori.json). The steps that do not need the user's judgement can also be run in one go, with $ authcat batch Arkalochori (Section 2.3).

5. References

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